Inheritance, Variation and Evolution

Complete some of the tasks from the grid below to reach a total of _____ points over this unit of work. Try and cover a variety of tasks over the unit so that you're practising different skills. Once you've completed a task, colour that box on the grid to keep a record of your points. Can you get the highest point score this unit?

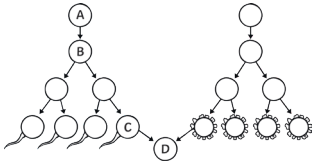
Topic	1 Point	2 Points	4 Points	6 Points	10 Points
Reproduction and Meiosis	Describe an organism that reproduces both sexually and asexually.	Explain the advantages of sexual and asexual reproduction.	Compare sexual and asexual reproduction.	Pick an animal and find out how many chromosomes are contained in each of its cells. Draw a diagram that shows how gametes are formed and fertilisation occurs. Label the number of chromosomes in each cell type.	Produce a video that explains the difference between mitosis and meiosis.
DNA and the Genome	State two reasons why it is important to study the human genome.	Describe the structure of DNA.	Make a model that shows where DNA is found in a cell, using whatever materials you'd like.	Write a newspaper article to persuade people that it is important to invest money in studying the human genome.	Research the scientist Rosalind Franklin. Explain why many people are dissapointed that she didn't receive a Nobel prize for her work.
Protein Synthesis and Mutations	Describe the difference between coding and non-coding DNA.	Name the parts of the cell that are involved in protein synthesis. Describe how they are involved.	Explain how proteins are formed from a sequence of DNA.	Make a comic strip that shows how protein synthesis occurs in a cell.	Explain, using examples, how a mutation in the DNA may result in a change in protein.
Inheritance	Describe the difference between genotype and phenotype.	Draw a diagram to show the relationship between genes and chromosomes.	Create a revision resource that will help you to learn the keywords for this topic.	If the **d** allele results in dimples and the **D** allele does not, draw a genetic diagram to show the inheritance of dimples in the children of a woman with dimples and a man who is heterozygous for the dimples allele.	Find out how Huntington disease is inherited. Produce a genetic diagram to explain the probability of a heterozygous dad with the disease and a mum without the disease, having children who inherit it

Genetic Disorders and Sex Determination	State how cystic fibrosis and polydactyly are inherited.	Explain how your sex was determined.	Give two reasons for and two reasons against embryo screening.	Write an information guide for parents who are considering embryo screening that tells them the benefits and risks associated with the procedure.	A couple are undergoing IVF treatment. They already have a child with sickle cell anaemia which is a painful disorder that shows recessive inheritance. Evaluate the use of embryo screening for this couple.
Variation	Draw a Venn diagram for genetic and environmental variation. Fill it with examples of variations between individual humans.	Explain what causes new variations to arise.	Produce a Facebook profile page for Darwin whilst he's on his voyage. Which locations does he check in to? What does he post about, and who comments?	Research some other species that Darwin recorded on his voyage. What genetic and environmental variations might he have found between individuals of the species?	Collect data on a variable (e.g. height, eye colour) from your friends and family. Plot your data on an appropriate graph.
Evolution	What is meant by the term 'natural selection'?	Write definitions for the keywords used during this lesson.	Use the development of Darwin's theory to explain how scientific theories develop over time.	Use Darwin's theory of natural selection to explain how the elephant's trunk evolved.	Produce a comic strip that shows how Darwin's theory of evolution by natural selection works.
Speciation	What is meant by the term 'species'?	Write at least three tweets directed at Darwin, at the time that he published his results, from people who object to his ideas.	Use Lamarck's theory to explain how the elephant's trunk evolved.	Explain how the different species of Darwin's finches could have developed from an ancestral species of finch.	Write a letter from Wallace to Darwin that describes some of the evidence he has collected and explains his ideas on speciation.
Evidence For Evolution: Fossils	What can we learn from fossils?	Explain why scientists can't be certain how life began.	Use your knowledge from physics to explain how scientists are able to date fossils.	Produce a poster aimed at primary school children that explains the different ways in which fossils might form.	Research the fossil record of a different organism to the ones we studied in class. Explain how the species has changed over time and what might have contributed to these changes.

Evidence For Evolution: Resistant Bacteria	What caused the rise in MRSA?	Write a patient leaflet to be displayed in doctors' waiting rooms. It should explain why doctors might not prescribe antibiotics and what patients should do if they are given a course of antibiotics.	Produce a storyboard that shows how bacteria develop antibiotic resistance.	Penicillin is an antibiotic. It was used a lot in the past to treat infections but there are now penicillin-resistant strains of bacteria. Explain how natural selection could have produced these strains of penicillin-resistant bacteria.	Produce a video or animation that shows how antibiotic resistance develops in bacteria.
Understanding of Genetics	The sweet pea produces lots of seeds. Why does this make it a good model?	Draw a punnet square to explain how two heterozygous parents produce offspring with a ratio of 3:1.	Explain how scientific theories develop over time.	Describe Mendel's experiments and explain how these led to his model of inheritance.	Produce a comic strip timeline that shows the discoveries that have led to our current understanding of genetics.
Selective Breeding	What is selective breeding?	Describe the steps involved in selective breeding.	Write a social media post about the positive or negative side of selective breeding. Then add a comment on the post that provides the opposite view.	Conduct your own research into selective breeding. Find one example of selective breeding for good, and one example of selective breeding gone wrong. Explain the case in your own words, highlighting the advantages/ disadvantages. You can present this information in any way you choose.	Make a video or animation that shows the process of selective breeding.
Genetic Engineering	What is genetic engineering?	Give some uses of genetic engineering.	Research a use of genetic engineering that was not covered in the lesson.	Produce an information leaflet that informs the public how genetic engineering is used in agriculture.	Make a film or write a script that describes the advantages and disadvantages of genetic engineering.

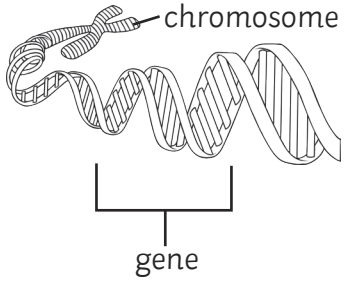
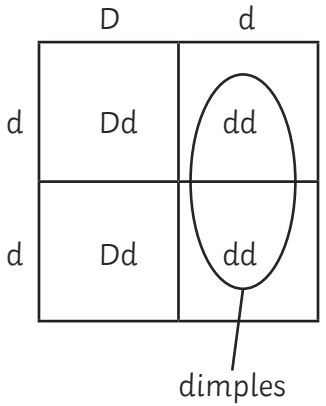
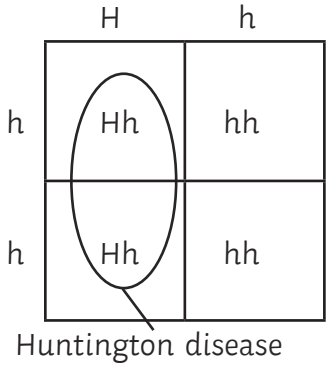
Cloning	Name the four methods of cloning.	Describe the different methods of cloning plants.	Give two benefits and two risks of cloning.	Compare and contrast the four methods of cloning.	Research the cloning of Dolly the sheep and produce a news article that explains the procedure used, the benefits of the research and any concerns raised.
Classification	Describe how organisms are named and give an example.	Describe how Linnaeus classified living things and give an example.	Draw a diagram that shows the three-domain system of classification, including the kingdoms in each. Give an example of an organism that would fit in each kingdom and explain why.	Produce a presentation that explains how evolutionary trees are used by scientists to show how organisms are related. Include the different information that could be used to produce a tree and how this might affect the shape of the tree. Explain what kind of data is used to place extinct animals into a tree. Include examples of some relationships.	Use your knowledge from the whole unit to explain, with examples, how scientific methods and theories develop over time.

Inheritance, Variation and Evolution Answers

Topic	1 Point	2 Points	4 Points	6 Points	10 Points
Reproduction and Meiosis	daffodil - bulbs divide strawberry - runners malaria - sexual reproduction in the mosquito and asexual in the human fungi - Releases spores which are produced asexually, or uses hyphae to sexually reproduce with another fungi if conditions are not good.	Sexual reproduction produces variation in the offspring which helps them to survive environmental changes by natural selection. Asexual reproduction only requires one parent which means it is more time and energy efficient, since they don't need to search for a mate. It is faster than sexual reproduction. Asexual reproduction produces many clones when the conditions are favourable.	Sexual reproduction involves two parents, whilst asexual reproduction only needs one. Sexual reproduction involves gametes which are produced by meiosis. There is no fusion of gametes in asexual reproduction. Only mitosis is involved in asexual reproduction. Asexual reproduction produces genetically identical offspring (clones). However, there is mixing of genetic information in sexual reproduction that causes variation in the offspring.	The chromosome numbers will depend on the animal the student has chosen, but you're looking for them correctly identifying which cells will have half the number of chromosomes. The diagram should look similar to the following.  	Meiosis occurs in the ovary or testes and produces gametes for sexual reproduction. Mitosis occurs in all other cells - it is asexual reproduction that is used for growth and replacement of cells. Meiosis forms 4 daughter cells, in two divisions, and each has 23 chromosomes. Mitosis forms two daughter cells, in one division, and each has 46 chromosomes. Genetic information is not mixed in mitosis, so daughter cells are identical. In meiosis, there is mixing of the genetic information that causes variation in the daughter cells.

DNA and the Genome	**Two of: It helps us to search for genes that are linked to different types of disease.** **It helps us to understand and treat inherited disorders.** **We can use the information to trace human migration patterns from the past.**	**DNA is polymer made up of nucleotides. Each nucleotide is made of a sugar and phosphate group, with one of four different bases attached. The four bases are A, C, T and G. Two strands of DNA twist together to form a double helix. The two strands are complementary, an A base on one strand matches a T base on the other strand, while a C base matches a G base.**	**This will vary, depending on students choice of materials, but they should have found a way of showing that DNA is found in chromosomes and these are found in the nucleus.**	**The article should include the following points:** **It helps us to search for genes that are linked to different types of disease.** **It helps us to understand and treat inherited disorders.** **We can use the information to trace human migration patterns from the past.** **The best artices will use persuasive language and may give examples to engage the reader, such as examples of inherited diseases and the impact they can have.**	**Many people can name Watson and Crick as the scientists who modelled the structure of DNA, however they are less likely to have heard of Rosalind Franklin. It was her work using x-rays to study DNA that led Watson and Crick to complete their model. Many people felt she should have shared the Nobel prize for the discovery. Students may also refer to the pioneering work that she did on the tobacco mosaic virus and polio.**

Protein Synthesis and Mutations	**Coding DNA codes for a protein. Each protein is coded for by one gene. Non-coding DNA does not code for a protein. It often has a role in regulating gene expression.**	**The nucleus contains the DNA which codes for the protein. The ribosomes attach to the template and bring in the carrier molecules that form the amino acid chain. The mitochondria provide the energy required for protein synthesis.**	**A template of the DNA is made which travels to the cytoplasm. Here, it attaches to a ribosome, where carrier molecules attach to each three base pairs of the template. The carrier molecules bring one amino acid per three base code and these are joined together to form a chain of amino acids. The amino acid chain detaches from the ribosome and folds up to form the final protein shape.**	**The comic should show the following:** **A template of the DNA is made which travels to the cytoplasm. Here, it attaches to a ribosome, where carrier molecules attach to each three base pairs of the template. The carrier molecules bring one amino acid per three base code and these are joined together to form a chain of amino acids. The amino acid chain detaches from the ribosome and folds up to form the final protein shape.**	**Each set of three bases codes for an amino acid. If one of the bases changes, then it might cause a change in the amino acid. If the sequence of amino acid in the chain changes, then the protein may fold differently and result in a different shape. This shape may not be able to carry out its original role in the same way. Students may include examples of inherited disorders or just changes in characteristics - they don't need to have included the actual mutations, just an idea of how changing the protein may effect the phenotype.**

Inheritance	**The genotype is the combination of alleles that someone inherits and the phenotype is the trait that is expressed as a result of the genotype.**		**Students should have included the keywords: gene, allele, heterozygous, homozygous, dominant, recessive, genotype and phenotype.** **The best examples will include a creative way to record the keywords and definitions.**		**Huntington disease shows dominant inheritance.** **Male genotype: Hh. Female genotype: hh.** 
Genetic Disorders and Sex Determination	**Cystic fibrosis inheritance is recessive. Polydactyly inheritance is dominant.**	**The student received an X chromosome from their mum. If they are female they received an X chromosome from their dad and if they are male they received a Y chromosome from their dad.**	**A list of reasons for and against is provided in lesson 4, Genetic Disorders.**	**Students may pick different issues to cover but the information should be balanced. The guide should definitely include the risk of miscarriage and the benefits of being able to make an informed decision. The best leaflets will discuss the emotional impact of testing on the family.**	**Only healthy embryos would be inserted into the womb. This way, the parents would have a high chance of preventing their next child having the disorder.** **Spare embryos would be destroyed, some people disagree with destroying this potential life.**

					It would reduce the incidence of the disease. The procedure is expensive, but it would save long-term health costs of a child born with the disorder.
Variation	Variation such as scars and tattoos in the environment section. Variation like eye colour and tongue-rolling is genetic. It could be argued that the majority of variations are affected by both, e.g. natural skin colour is genetic, but affected by sun exposure.	Random mutations in the DNA can cause a change in a species over a long time, if the new phenotype is useful in the environment.	The best examples would show Darwin sharing examples of species he has discovered and why he thought they were interesting. Students might have him check in at the location those species were found, or have other people ask questions about his discoveries.	Darwin found many species on his travels, including some large fossils and many different types of insects. Students should suggest sensible variations that would be present between individuals and decide if they are genetic, environmental or both.	Continuous data should be plotted on a scatter graph while discrete data should be plotted on a bar chart.

Evolution	Natural Selection: the process by which the individuals that are more able to compete can survive to reproduce.	Variation: differences in traits between individuals of a population. Competition: individuals in a population attempt to gain a share in the available resources. Adaptations: a trait that is better suited to the environment. Natural Selection: the process by which the individuals that are more able to compete can survive to reproduce. Inheritance: Alleles are passed on from parent to offspring.	Darwin collected a lot of evidence that helped him form and adapt his theory of evolution. He collaborated with many other scientists who gave him further evidence and ideas to develop his theory into the one he eventually published.	There was variation in trunk length. The elephants competed for food and water, and those with the longer trunks were better able to compete. These elephants were more likely to survive and reproduce - passing the alleles for their long trunk to their offspring.	The best comics would include: variation between individuals in a trait, competition for a resource, some of the individuals surviving whilst others don't, reproduction and offspring carrying the same trait as the parents.

Speciation	Organisms that can successfully breed together.	The tweets should cover: not having enough evidence; not being able to explain how inheritance happens; and the theory challenges religious beliefs.	The elephants stretched their noses during their lifetime to reach food and passed on their longer noses to their offspring.	The ancestral species became isolated on the different islands. The islands had different environments and so different variations will have been beneficial on different islands. For example, where nuts were abundant, species with large beaks were more likely to compete for food and therefore survive and reproduce. They passed their genes for larger beaks to their offspring. Eventually, the population became so different, they couldn't interbreed to produce fertile offspring with species on other islands.	Wallace observed a clear dividing line between species on his travels through Indonesia. Students may research and find he did a lot of his work on speciation, looking at warning colouration. They should explain how there is some form of environmental isolation that leads to an ancestral species developing in two different environments. Natural selection occurs and eventually a new species arises when it can no longer successfully interbreed with the other species.

Evidence For Evolution: Fossils	We can learn how much or how little different organisms have changed as life has developed on Earth.	Earliest life forms were soft-bodied so they decayed. The conditions needed for fossils to form are rare. Geological activity has destroyed a lot of fossils. We haven't discovered all the fossils yet.	Scientists measure radioactive decay from the rocks around the fossils. As they know the rate at which certain elements decay, they are able to determine how old the rocks are.	Students should have clear, simple explanations with diagrams that show: 1) conditions for decay are absent (amber, peat, ice); 2) minerals replace the hard parts of the organism; 3) traces of organisms are preserved.	This task will vary, depending on the organism that the student picks. They should have identified changes in the fossil record that show how the species has evolved, and suggestions about why they might have evolved in this way.

Evidence For Evolution: Resistant Bacteria	**Doctors overused antibiotics. Patients didn't finish their courses of antibiotics. There was an overuse of antibiotics in agriculture.**	**Your doctor will not prescribe antibiotics if you have a virus because antibiotics only work on bacterial infections. Your doctor will not prescribe antibiotics if your infection is not serious and he/she thinks your body will be able to fight the infection itself. If you are given antibiotics, it's very important you complete the course so that there is less opportunity for resistant strains of bacteria to develop.**	**A storyboard that shows multiple bacteria with a random mutation in one bacterium that gives it antibiotic resistance. The use of an antibiotic that kills the other strain but that the resistant strain is more able to compete, survive and multiply. A further random mutation may give resistance to another antibiotic.**	**A random mutation in the bacteria gave resistance to penicillin. Therefore, some bacteria survive the penicillin and these reproduce by binary fission. The gene for penicillin resistance is passed on to the offspring.**	**An animation or video that shows multiple bacteria. A random mutation in one bacterium that gives antibiotic resistance. The use of an antibiotic that kills the other strain but that the resistant strain is more able to compete, survive and multiply. A further random mutation may give resistance to another antibiotic.**

Understanding of Genetics	**Using a sweet pea pod, you can repeat an experiment many times, producing lots of data. This means that you could better determine average values and identify any anomalous results more easily.**	<table><tr><td></td><td>R</td><td>r</td></tr><tr><td>R</td><td>RR</td><td>Rr</td></tr><tr><td>r</td><td>Rr</td><td>rr</td></tr></table> **There is a ratio of 3 dominant traits to 1 recessive.**	**There are usually lots of scientists who contribute evidence to a theory.** **The ideas are adapted gradually as new evidence is discovered.** **It can take many years and lots of evidence before a new theory is accepted.** **New technology can often contribute to the discovery of new evidence.**	**Mendel made pure breeding lines for each of the characteristics he investigated. When he bred these together, only the dominant trait was observed in the offspring. In the second generation, he allowed the plants to self-fertilise and these produced offspring in a 3:1 ratio of dominant to recessive traits. He concluded that characteristics are controlled by pairs of heritable 'units' that come in different versions. The dominant unit can mask the presence of the recessive unit. The two paired units separate during gamete production, which means that each egg or sperm only carries one unit.**	**The comic strip should include: Mendel and his pea experiments, the chromosomes dance of mitosis, the observation of meiosis and the discovery of the structure of DNA.**

Selective Breeding	**Humans control selection by choosing organisms to breed that have desired traits.**	**1. Choose the plants or animals from your existing stock that have the feature you want.** **2. Breed these with each other.** **3. Choose the best of the offspring and breed these together.** **4. Continue this process over several generations.** **5. Eventually, all the offspring have the features that you are looking for.**	**This will vary, depending on the student's social media post and the choice of argument. However, you're looking for an argument with supporting evidence either for or against selective breeding, with a comment to balance the argument.**	**This will depend on the example that the student finds. You're looking for a balanced argument that gives both advantages and disadvantages of selective breeding, backed up by evidence from their research. This shouldn't be an example they studied in class.**	**You're looking for human involvement in selection - breeding of two organisms with the desired trait and then selection of the offspring with examples of this trait. The student should have shown that this repeats over many generations.**
Genetic Engineering	**Genetic engineering is a process that involves modifying the genome of an organism by introducing a gene from another organism to give a desired characteristic.**	**Golden rice to reduce vitamin A deficiency. Producing insulin for diabetics. GM crops that produce a larger yield or grow in more difficult climates. Animal models of disease.**	**This will depend on what the students research.**	**The leaflet should cover information about GM crops and the advantages of engineering them, as well as the concerns people may have.**	**Possible advantages:** **• GM crops have bigger yields and therefore have increased food value. This may help us to address an expanding population.**

					• Animals and plants grow faster. • We can mass produce human proteins. • GM crops can grow in dry, hot, cold or flooded parts of the world. This may help us to manage the effects of global warming on agriculture. • GM crops can produce their own pesticide, can be resistant to herbicides, or can contain specific nutrients. • Models of human diseases can help us to develop treatments.

					Possible disadvantages: • **Nobody can be sure of the long-term effects of eating GM food on human health.** • **Pesticide-producing crops may result in insects developing resistance to pesticides.** • **Genes from GM crops or animals could spread to species in the wild.** • **People worry that this may lead to the manipulation of human genes to make 'designer' children.**

Cloning	**Cuttings, tissue culture, embryo transplants, adult cell cloning**	**Taking cuttings involves taking a newly-growing stem and adding rooting powder to encourage root formation to grow a clone of the plant. Tissue culture takes a small part of the plant tissue and stimulates growth in agar using different hormones. This method can produce many plants from a small amount of tissue.**	**Ability to make clones of genetically modified plants or animals that are useful medically or nutritionally for humans.** **Risks: Some people fear this may lead to the cloning of humans.** **The techniques produce lots of organisms with identical genes. This reduces variation in the population which means they may be less able to survive any changes to the environment that may happen in the future.** **If a whole herd of cattle, or field of crops, is cloned and catches a disease they are susceptible to then the whole population may be wiped out in one go.**	**Things students could mention:** **Whether the techniques are for plant or animal cloning.** **Whether they involve adult or embryonic cells.** **How reliable the method is.** **The benefits of the method.** **The impact the method could have in the future.**	**Students should explain how adult cell cloning works with reference to Dolly the sheep. The best examples will include information about the reliability of the technique at the time. They should include the benefits of this technological advance and balance this with concerns that have been raised about the research.**

			Many people argue that carrying out these processes on animals is not ethical. The animals have no say over these procedures being carried out on them, in some cases we are forcing pregnancy on them, or injecting them with hormones and harvesting their eggs. Adult cell cloning is not very reliable, and this means many hundreds of embryos die in the process.		

Classification	**Binomial naming system of genus and then species, with an example of a binomial name for an organism.**	**Kingdom, phylum, class, order, family, genus, species and an example of an organism and which of each it belongs to.**	**The diagram should include: archaea, with the kingdom archaeabacteria; bacteria, with the kingdom eubacteria; and eukaryota with the four kingdoms: protists, fungi, plants and animals.** **Students may use an explanation like, 'the cells don't have a nucleus and they are found in extreme conditions'.**	**This should explain how morphology can be compared to look for similarities between organisms, but that DNA and protein sequencing is more accurate and can show different relationships. Fossil data will be used to show where extinct animals may have fitted into an evoutionary tree. The best examples will include illustrations of evolutionary trees and information about relationships that students have researched.**	**Students could include information about microscopes and DNA sequencing changing the way in which organims are classified. Microscope methods allowing scientists to see chromosomes in cells and observe their behaviour in mitosis and meiosis, therefore confirming Mendel's ideas on units of inheritance. X-rays being used to show the structure of DNA. More data on speciation being used to support Darwin's theory of evolution by natural selection.**